Lab Report

Summer 2025

Course Code: **EEE203L**

Course Title: **Electronic Circuits II Laboratory**

Lab Section:**02**

Experiment No.: 6

Name of the experiment: Familiarization with Series resonance in AC Circuits

Date of Submission:

Group No: **3**

Group Member Details:

1.	*Name:*	**Souvik Barman Ratul**
	I.D.	**24121205**
2.	*Name:*	**Oshiul Alim Ornob**
	I.D.	**24121268**
3.	*Name:*	**Mahdi Zaman**
	I.D.	**24121307**
4.	*Name:*	**Alif Tamjid**
	I.D.	**24121308**

Submitted By :

Name : Souvik Barman Ratul

I.D. : 24121205

Submitted To :

Saad Mahbub Chowdhury

Objective:

The purpose of this experiment is to understand how resonance occurs in a series resonance occurs in a series RLC circuit and what its effects are on current, voltage and power.

Theoretical Background:

When a resistor, inductor and capacitor are connected in series with AC source, the inductor opposes current and the capacitor delay currents. At a special frequency called resonant frequency, the inductors effect cancels out the capacitors effect ($X_L = X_C$)

→ The impedance is minimum, equal to R

→ The current is maximum.

→ V and I are in phase

→ The circuit absorbs maximum power.

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Here, resonance makes the circuit highly efficient by allowing maximum current and power transfer.

Compilation of result and Calculations (with ckt diagram and verification of result):

Circuit Diagram:

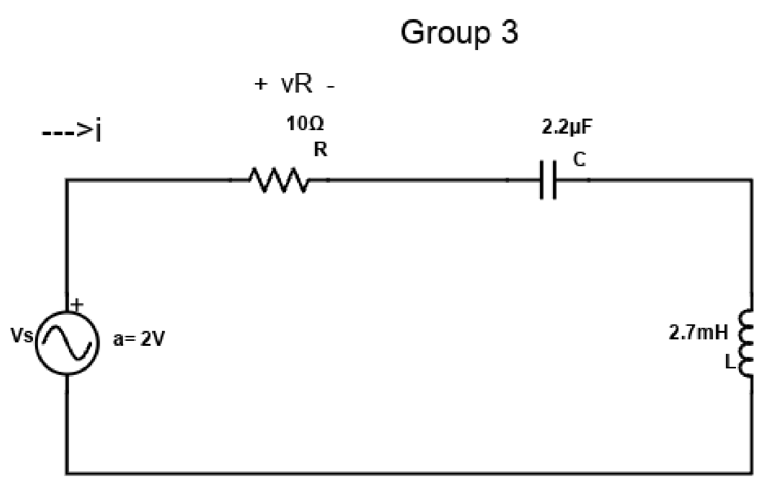


fig. Experiment 6

Data Table:

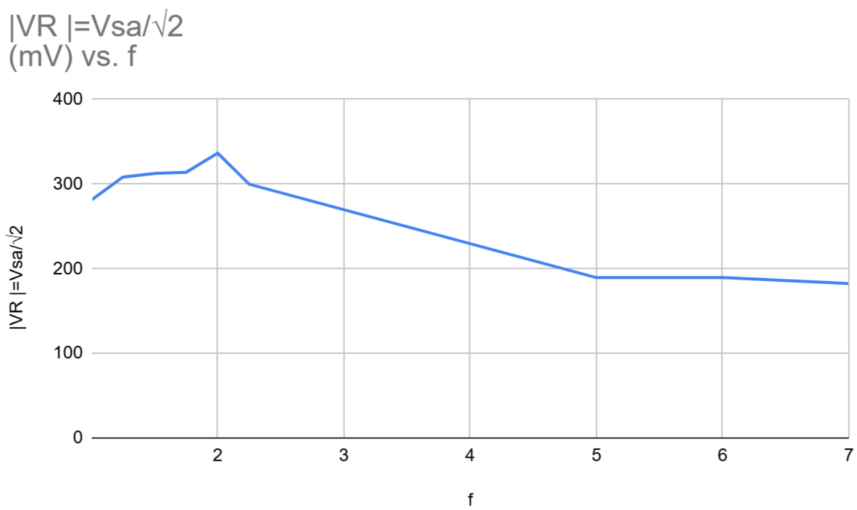
R (Ω)	C (μF)	L (mH)	$fst=1/2\pi\sqrt{CL}$ (KHz)	Vsa (V)	fsp (kHz)	$\|Vs\|=Vsa/\sqrt{2}$ (V)	$\angle Vs$ (0)
10.8	2.19	2.66	2.085	2.04	1.003	1.7	0

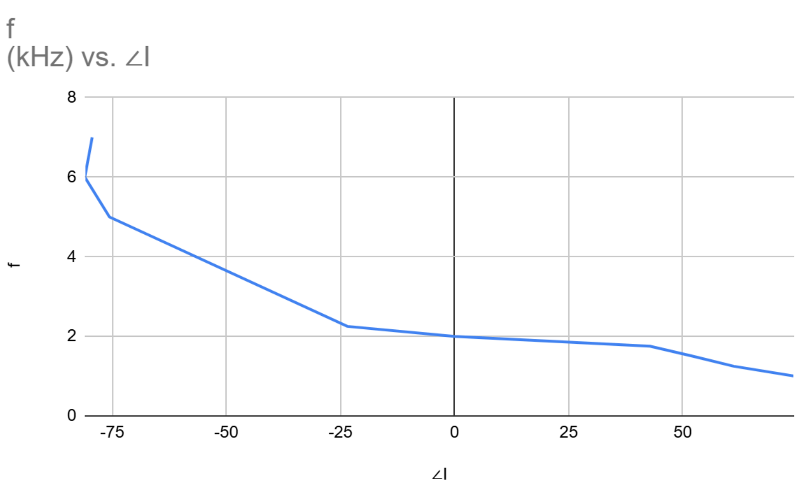
f (kHz)	Vsa (V)	VRa (mV)	$\|V_R\|=Vsa/\sqrt{2}$ (mV)	$\|I\|=V_R/R$ (mA)	Sign (+/-)	Δt(ms) (From oscilloscope)	$360f\Delta t$ (o)	$\angle I$ (0)
1.003	2.04	398	281.43	26.058	+	0.206	74.38	+74.38
1.249	1.72	438	308.298	28355	+	0.136	61.151	+61.151
1.505	1.20	442	312.541	28.94	+	0.096	52.01	+52.01
1.75	0.90	444	313.96	29.07	+	0.068	42.84	+42.84
2	0.71	476	336.58	31.17	0	0	0	0
2.25	0.73	424	299.81	27.76	-	0.029	23.49	-23.49
5	1.94	268	189.5	17.55	-	0.042	75.6	-75.6
6	2.36	268	189.5	17.55	-	0.0375	81	-81
7	2.56	258	182.43	16.9	-	0.0315	79.39	-79.39

f (kHz)	$\|Z\|=\|Vs\|/\|I\|$ (KΩ)	$\angle Z=\angle Vs-\angle I(0)$
1.003	0.0554	-74.38
1.249	0.0426	-61.151
1.505	0.0293	-52.01

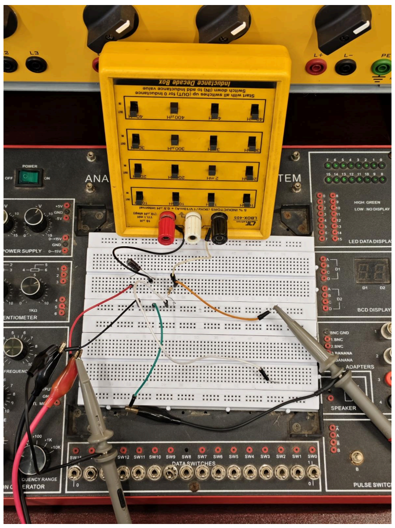
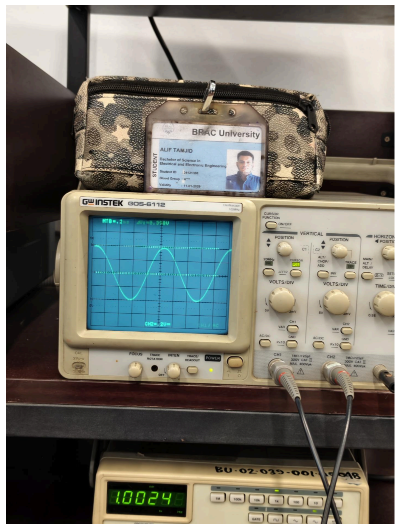
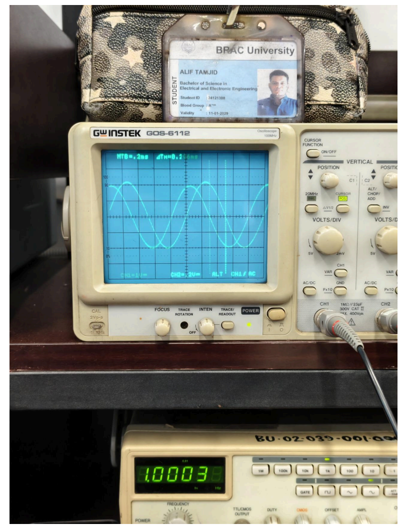
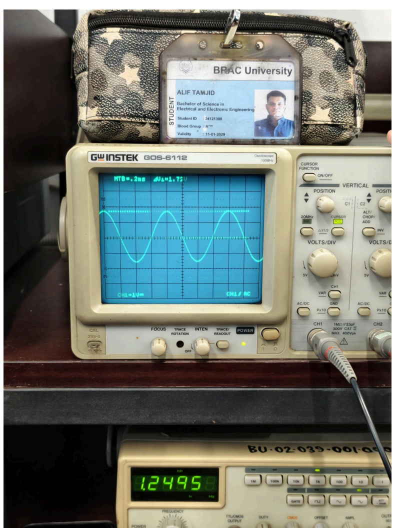
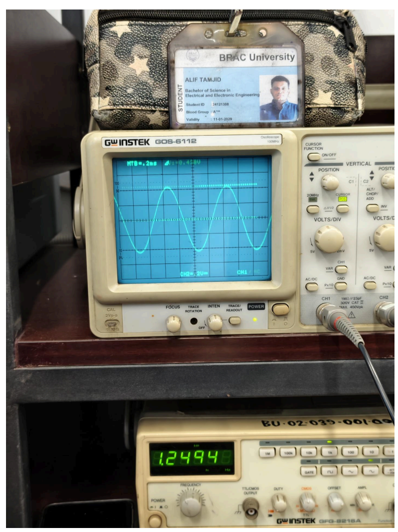
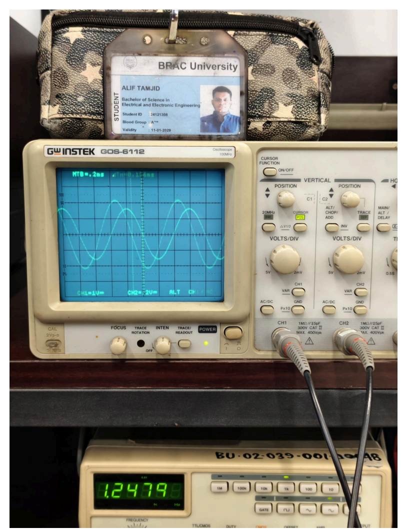
1.75	0.0219	-42.84
2	0.0161	0
2.25	0.186	+23.49
5	0.0782	+75.6
6	0.0951	+81
7	0.107	+79.39

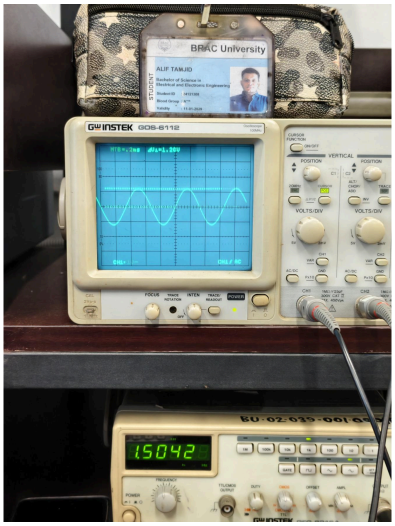
<u>Graph</u>





<u>Classwork Proof :</u>

Vsa (V)	V_{Ra} (mV)	Δt(ms)
		
		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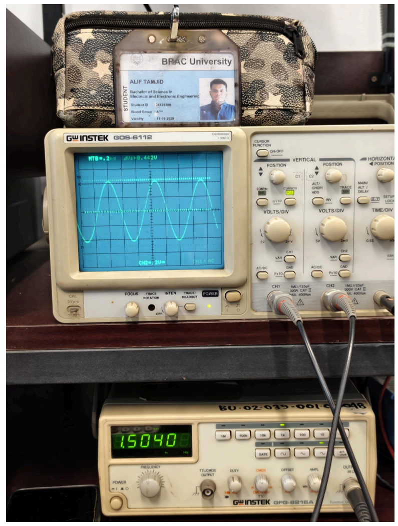
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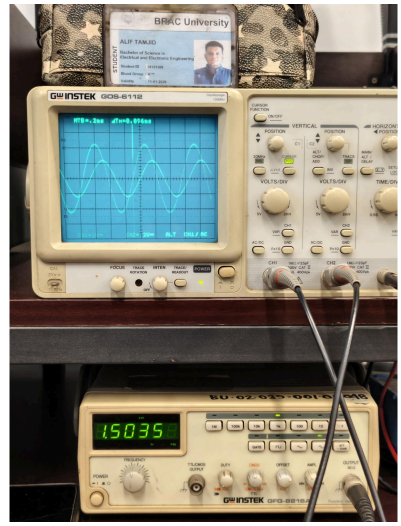
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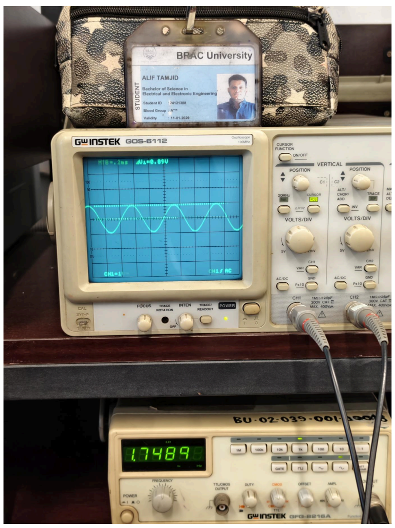
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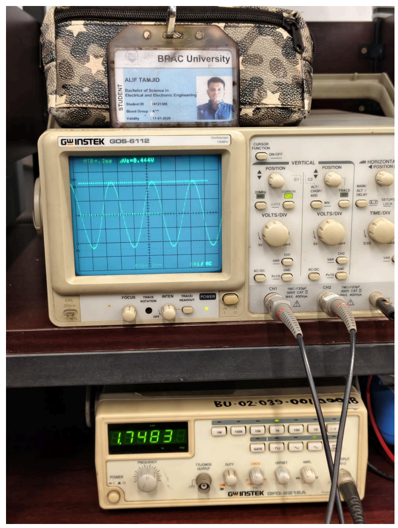
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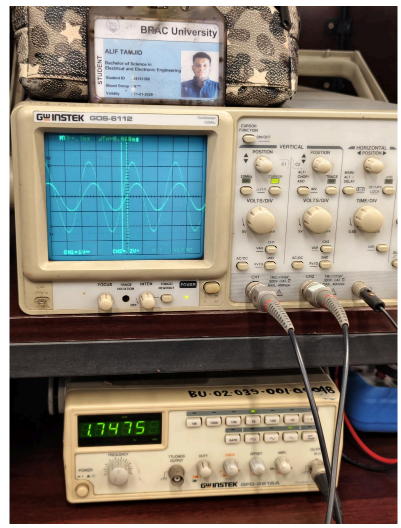
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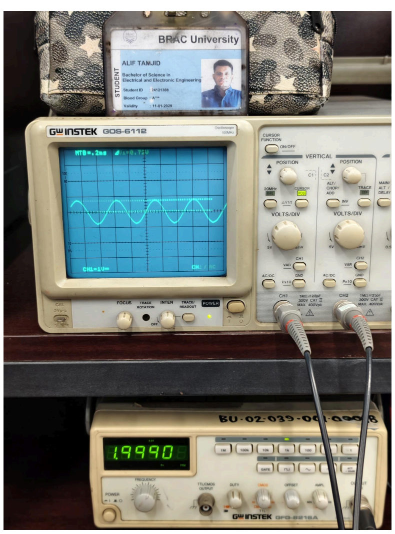
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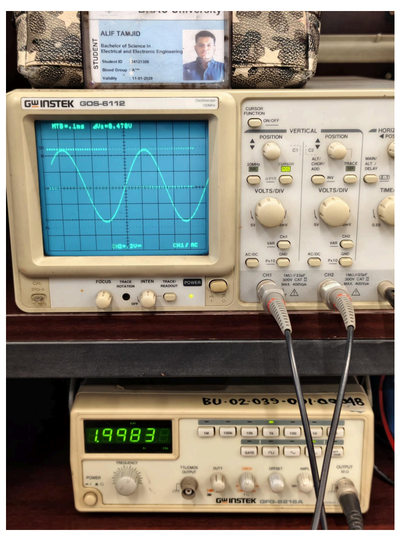
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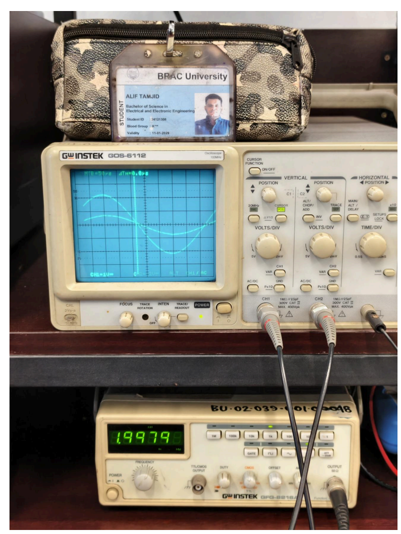
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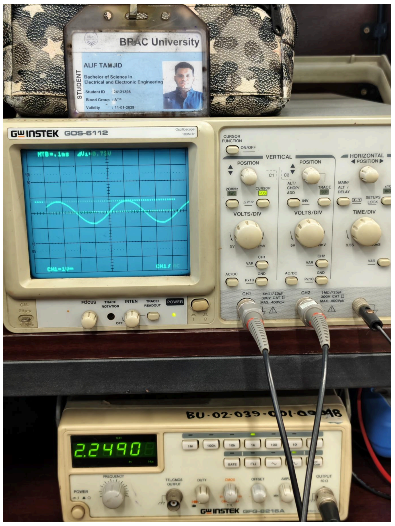
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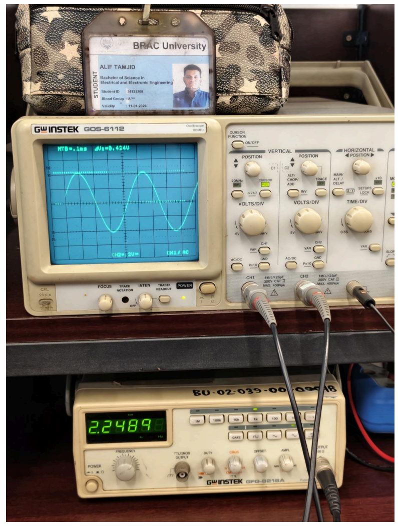
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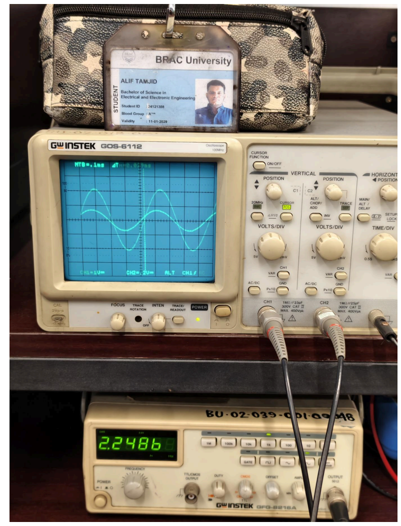
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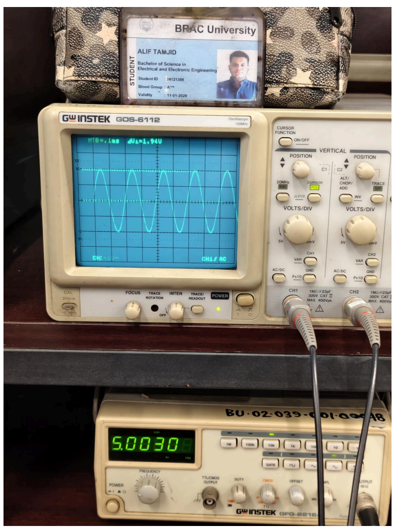
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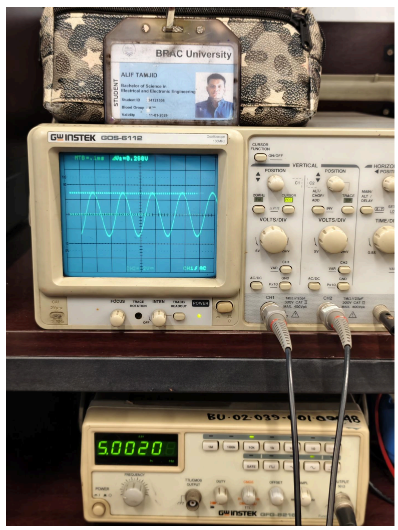
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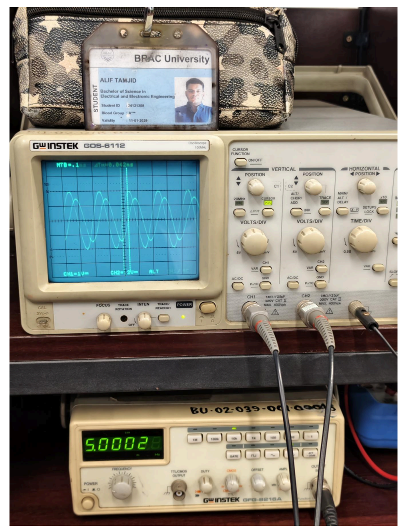
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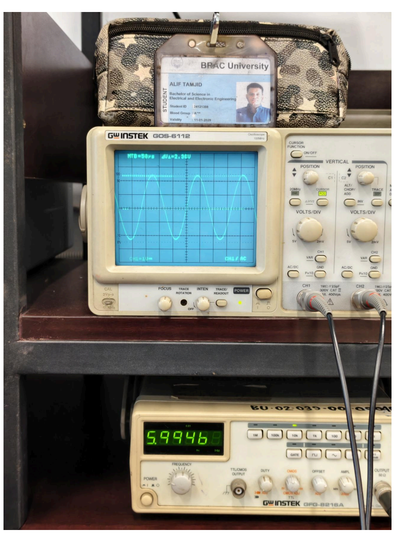
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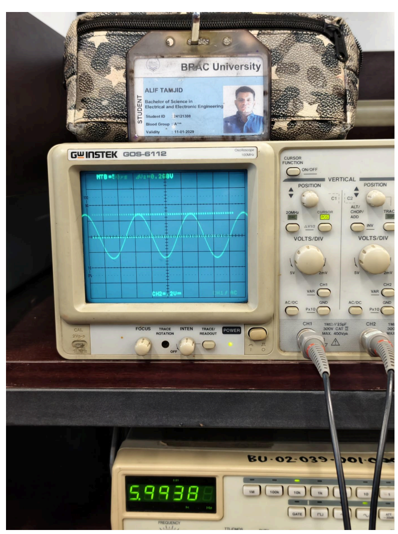
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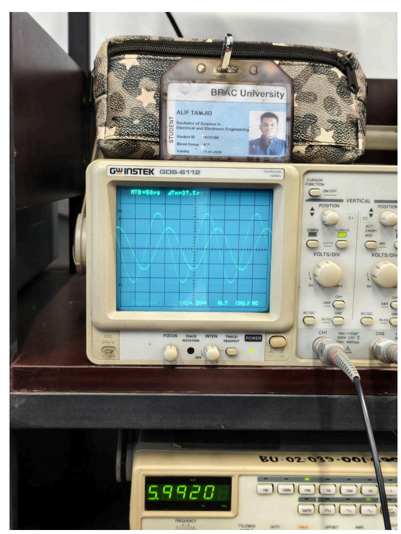
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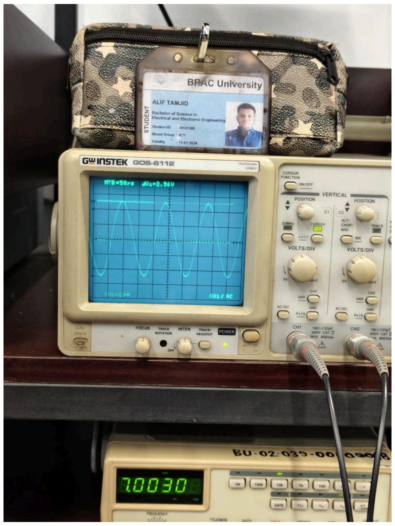
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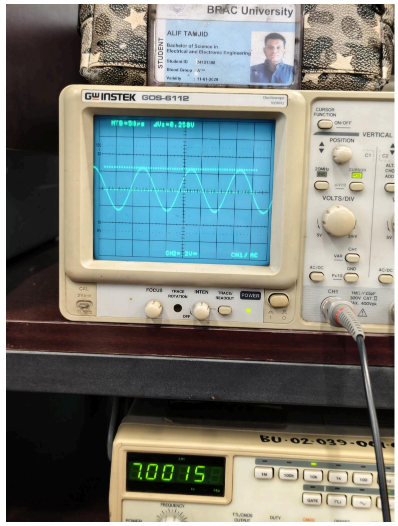
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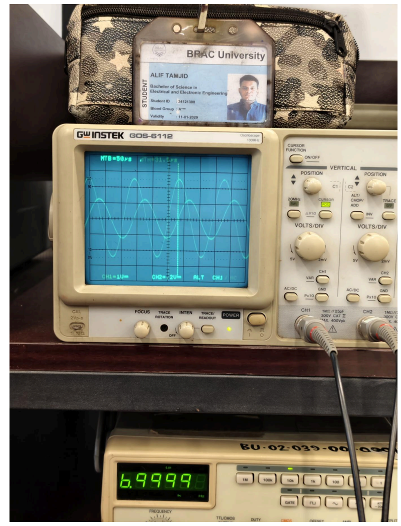
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Appendix:

Brac University
Department of Electrical & Electronic Engineering (EEE)
EEE203L – Electrical Circuits II Laboratory

Experiment 6

Name of the Experiment

Familiarization with Series Resonance in AC Circuits

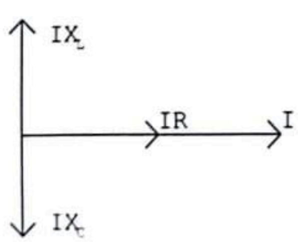
Objective:
This experiment helps students understand the resonance phenomenon in R-L-C series circuits.

Apparatus:

1. Resistor
2. Capacitor
3. Inductor
4. Signal Generator
5. Oscilloscope
6. Multimeter

Introduction:

Resonance is a particular situation that may occur in an electric circuit containing both inductive and capacitive elements. In a series RLC circuit resonance occurs when the resultant reactance is zero. We know, the drop across the inductance leads the current by 90^0 whereas that across the capacitor lags the current by 90^0. The two drops are opposite. If they are made equal as in figure below, the inductive and capacitive voltage drops neutralize each other. So, the total voltage drop is equal only to the resistive drop.



Inspecting the above figure it can be said that, during series resonance the applied voltage drop is in phase with current and the power factor is unity.

Thus for series resonance,

$IX_L = IX_C$

That concludes, Inductive Reactance = Capacitive Reactance

- ⇨ $X_L = X_C$
- ⇨ $2\pi fL = \frac{1}{2\pi fC}$ ....(1)

Therefore, the series resonant frequency is,

$f_r = \frac{1}{2\pi\sqrt{LC}}$ $= 2065.03\ Hz$

At resonant condition for a series circuit the following should be observed:

- The resultant impedance of the circuit is purely resistive in nature:

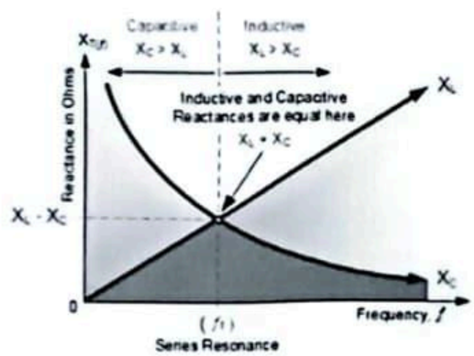
  $Z_T = R + j\,(X_L - X_C) = R + j.0 = R$

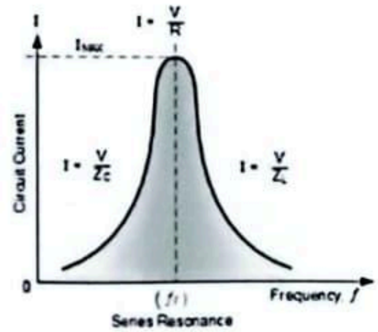
- Since the impedance is minimum the current in the series circuit will be maximum and given as: $I_{max} = \frac{V_s}{R}$
- The circuit yields a unity power factor since the impedance is purely resistive.
- The average power absorbed by the circuit is maximum at this resonance: $P_{max} = I_{max}^2\,R$

Practical considerations to be taken for the experiment:

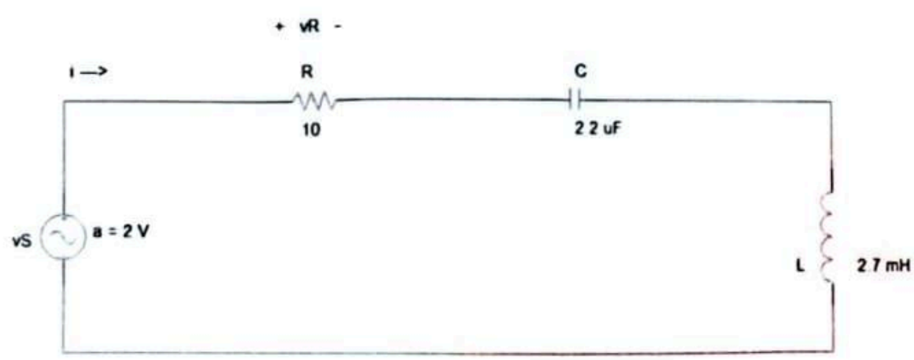
- Due to loading effect the amplitude of the source voltage changes when we use different capacitors. So one should check the amplitude every time a new capacitor is used & make adjustment if necessary. Remember that we should keep same amplitude value throughout the experiment.
- Since at resonance point the circuit current is maximum, the watt dissipation of the resistor R must be checked against its wattage rating. It should be applicable also for the inductor current rating. Besides the signal generator we are using must be capable of delivering maximum current.
- In a series circuit containing R, L & C the voltage across the C can raise beyond the supply voltage. So other circuit equipment those are subjected to VC must be able to withstand that voltage.
- From the current vs. capacitance curve, it should be noted that the current is maximum at resonance point and decreases beyond that. The shape of the curve (flatness) mostly depends on the value of 'R'. A large value of 'R' will give rise to a flat curve and it will make difficult for us to measure the variation of current at different capacitor values. So we should use a smaller value of 'R'.

- When measuring the phase angle between voltage & current we should check the connection of the oscilloscope probes carefully. Each of the black crocodile clip represents the ground of respective oscilloscope channel. But inside the oscilloscope these two grounds are shortened. So when we use both the probes there is a great chance that we short-circuit some parts of the circuit. We should always avoid this situation.





Circuit:



Procedure:

1. Connect the circuit as shown in the circuit diagram.
2. Adjust the voltage from the signal generator to 2 V (zero-peak).
3. Keep the frequency to some minimum value (say 10 Hz) and note the current.
4. Increase the frequency and for each frequency note the corresponding current.
5. Draw the resonant curve by taking (I) along y-axis and (f) on x-axis.

Data Table:

R (Ω)	C (μF)	L (mH)	$f_{ST} = \frac{1}{2\pi\sqrt{CL}}$ (kHz)	V_{Sa} (V)	f_{SP} (kHz)	$\lvert V_S \rvert = \frac{V_{Sa}}{\sqrt{2}}$ (V)	$\angle V_S$ (o)
10.8	2 19	2 16	2.085	2.04	1.003	1.72	0

For ①

f (kHz)	V_{Sa} (V)	V_{Ra} (mV)	$\lvert V_R \rvert = \frac{V_{Ra}}{\sqrt{2}}$ (mV)	$\lvert I \rvert = \frac{\lvert V_R \rvert}{R}$ (mA)	Sign (+/-)	Δt(ms) (From oscilloscope)	360fΔt (o)	∠I (o)
1.003	2.04	398	281.43	26.058	+	0.206	74.38	+74.38
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1.505	1.20	442	312.541	28.94	+	0.096	52.01	+52.01
1.75	0.90	444	313.96	29.07	+	0.068	42.84	+42.84
2	0.71	476	336.58	31.17	0	0	0	0
2.25	0.73	424	299.81	27.76	-	0.029	23.49	-23.49
5	1.94	268	189.5	17.55	-	0.042	75.6	-75.6
6	2.36	268	189.5	17.55	-	0.0375	81	-81
7	2.56	258	182.43	16.9		0.0315	79.9	-79.39

f (kHz)	$\lvert Z \rvert = \frac{\lvert V_S \rvert}{\lvert I \rvert}$ (kΩ)	∠Z=∠Vs - ∠I (o)	f (kHz)	$\lvert Z \rvert = \frac{\lvert V_S \rvert}{\lvert I \rvert}$ (kΩ)	∠Z = ∠V_S - ∠I (o)
1.003	0.0554	-74.38	2.25	0.186	+23.49
1.249	0.0426	-61.151	5	0.0782	+75.6
1.505	0.0293	-52.01	6	0.0951	+81
1.75	0.0219	-42.84	7	0.107	+79.39
2	0.0161	0	×	×	×

Questions:

1. Write the effects of series Resonance in AC circuit.
2. Draw necessary graphs for the RLC Series circuit.
3. What will happen to the plot of the current vs. frequency if the value of the Resistor (R) is varied?

Saad 12/08/25

Conclusion:

In this experiment, we saw that resonance happens in a series RLC circuit when the inductor and capacitor balance each other. At this point, the circuit acted like a pure resistor. The current became highest, and voltage and current were in the same phase. This proved that at resonance the circuit power factor is 1 and power transfer is maximum.

Lab Report

Summer 2025

Course Code: **EEE203L**

Course Title: **Electronic Circuits II Laboratory**

Lab Section:**02**

Experiment No.: 7

Name of the experiment: Cut-off Frequencies, Bandwidth & Selectivity

Date of Submission:

Group No: **3**

Group Member Details:

5.	*Name:*	**Souvik Barman Ratul**
	I.D.	**24121205**
6.	*Name:*	**Oshiul Alim Ornob**
	I.D.	**24121268**
7.	*Name:*	**Mahdi Zaman**
	I.D.	**24121307**
8.	*Name:*	**Alif Tamjid**
	I.D.	**24121308**

Submitted By :

Name : Souvik Barman Ratul

I.D. : 24121205

Submitted To :

Saad Mahbub Chowdhury

Object:

The goal of this experment is to measure the band-width and Q-factor of a series RLC circuit at resonance.

Theoretical Background:

At resonance, the circuit current is maximum, but as a frequency moves away from the resonant frequency, current drops. If we plot current against frequency, we get a peak at the resonance frequency.

The two frequencies where the falls to 70% of maximum are called cut off frequency. The difference between them called Bandwidth ($BW = f_{CH} - f_{CL}$)

The Quality factor (Q)

$$\therefore Q = \frac{f_{SP}}{BW}$$

→ A higher 'Q' means the resonance is sharp and the circuit is more selective to one frequency.

→ A lower 'Q' means resonance is broad and less selective.

Compilation of result and Calculations (with ckt diagram and verification of result):

Circuit Diagram:

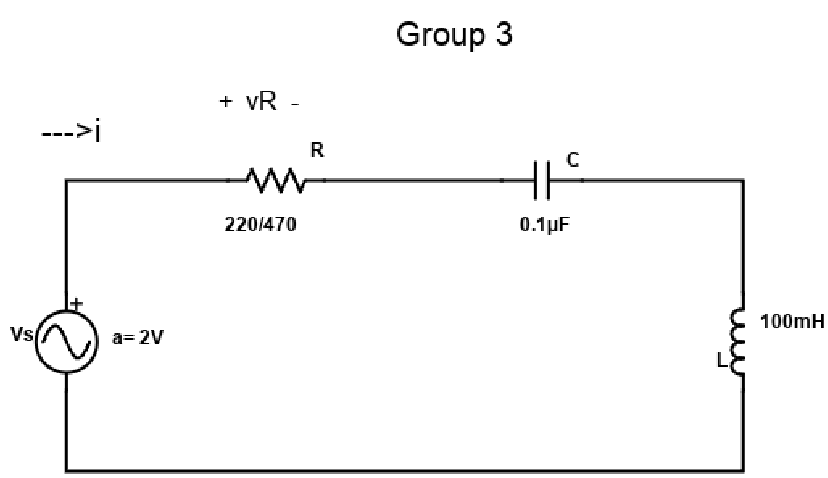
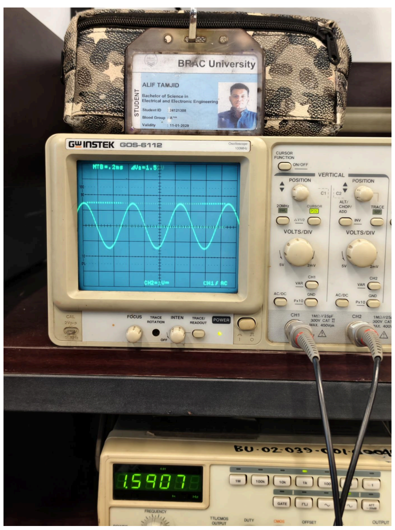
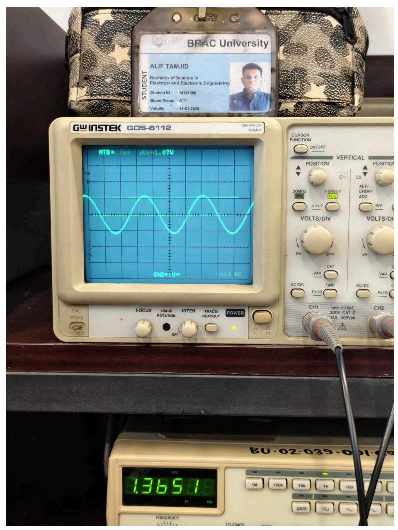
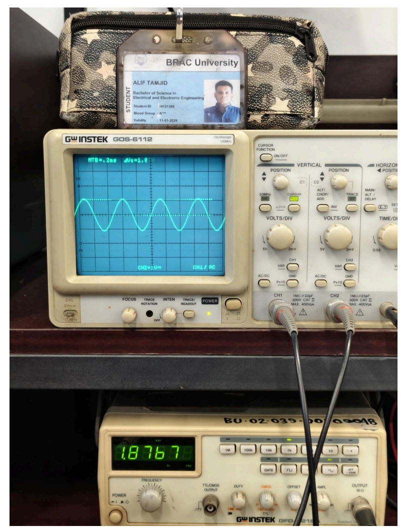
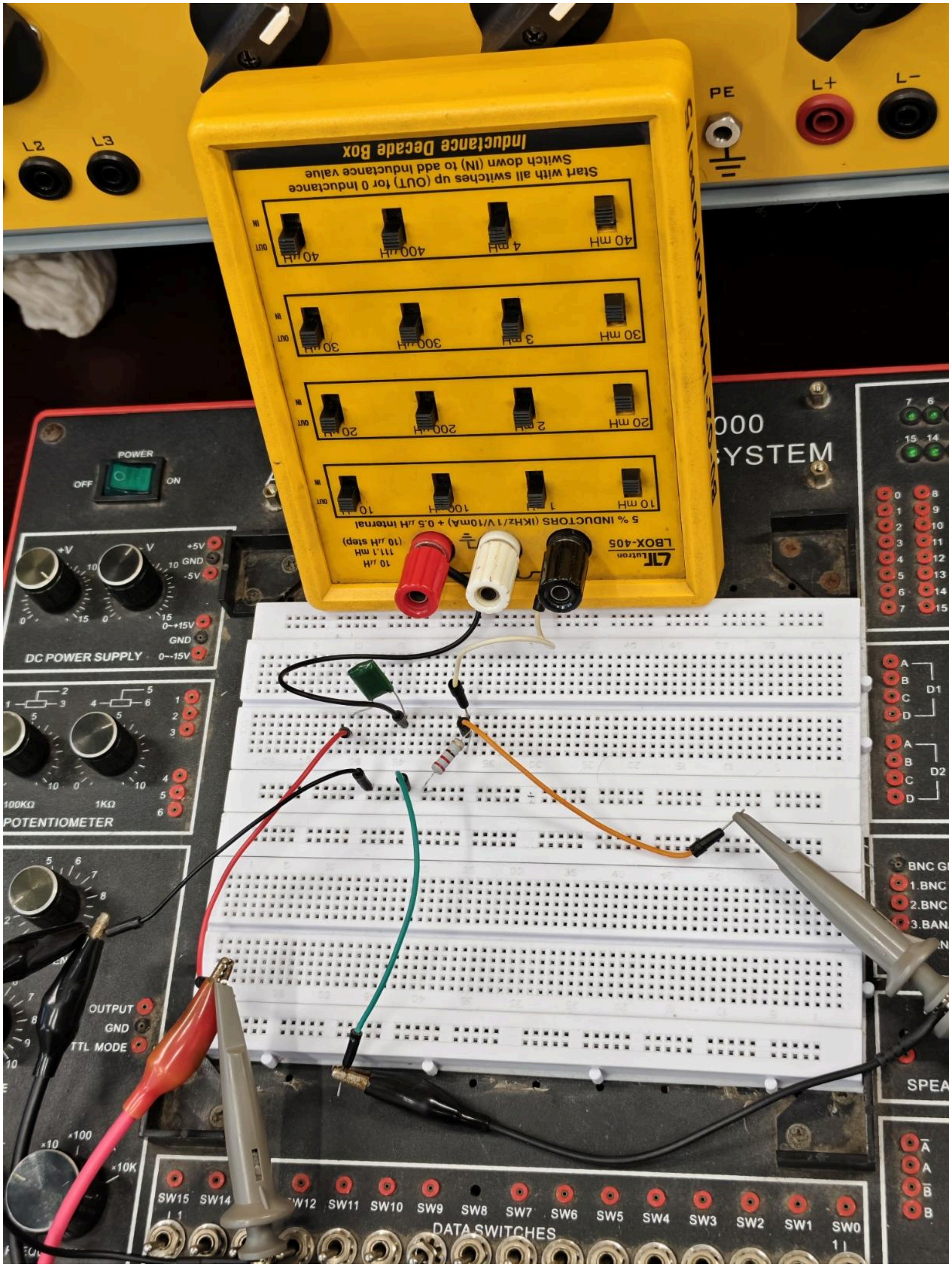


fig. Experiment 7

Classwork Proof:

V_{Ramax} (V) & fsp (kHz)	$0.707V_{Ramax}$ (V) & fcL (kHz)	$0.707V_{Ramax}$ (V) & fcH (kHz)
		


Inductance Decade Box
Start with all switches up (OUT) for 0 inductance
Switch down (IN) to add inductance value
LBOX-405
PE
L+
L−
L2
L3
POWER
OFF
ON
DC POWER SUPPLY
POTENTIOMETER
OUTPUT
GND
TTL MODE
DATA SWITCHES

Data Table:

C (μF)	L (mH)	$f_{st}=1/2\pi\sqrt{CL}$
0.1	100	1592Hz

R (Ω)	Vsa (V)	fsp (kHz)	V_{Ramax} (V)	$0.707V_{Ramax}$ (V)	Vsa (V)	f_{cL} (kHz)	Vsa (V)	f_{cH} (kHz)	$BW=f_{cH}-f_{cL}$ (kHz)	$Q_s=f_{sp}/BW$
220	2	1.59	1.52	1.075	2	1.37	2	1.87	0.5	3.18

Appendix:

Brac University
Department of Electrical & Electronic Engineering (EEE)
EEE203L – Electrical Circuits II Laboratory

Experiment 7

Name of the Experiment

Series Resonance: Cut-off Frequencies, Bandwidth & Selectivity

Objective:
Students are expected to find the Q-factor and Bandwidth of Series RLC circuit.

Apparatus:
1. Resistor
2. Capacitor
3. Inductor
4. Signal Generator
5. Oscilloscope
6. Multimeter

Introduction:

In a series R-L-C circuit, resonance can occur when

Inductive Reactance = Capacitive Reactance

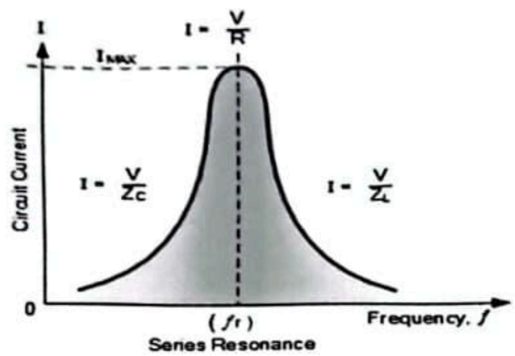
⇨ $X_L = X_C$

⇨ $2\pi fL = \dfrac{1}{2\pi fC}$ ....(1)

Thus it is possible to have resonance by varying the frequency 'f' while keeping L & C constant. For a series circuit, we should observe the followings while we vary 'f' in order to have resonance:

- The frequency at which the resonance occurs is known as resonant frequency 'f_r'. At this frequency, the current is maximum and the voltage drop across 'R' is also maximum.

Beyond this frequency, the current and as well as the V_R drops which gives rise to a profile as shown below:



- From the above profile, notice that the maximum value of voltage (representing the maximum current value) occurs at the resonant frequency 'f_r'.
- The half power point is defined as the point that corresponds to the points at which voltage is equal to $\frac{V_m}{\sqrt{2}} = 0.707Vm$.
- The frequencies (f_1 & f_2) corresponding to the half power points represents the cut-off frequencies for the circuit.
- The bandwidth of the circuit is defined as $BW = f_1 \sim f_2$.

In our experiment, we shall try to find the frequency response of the series R-L-C circuit and find its bandwidth. We shall conduct the following studies on quality factor of the series circuit:

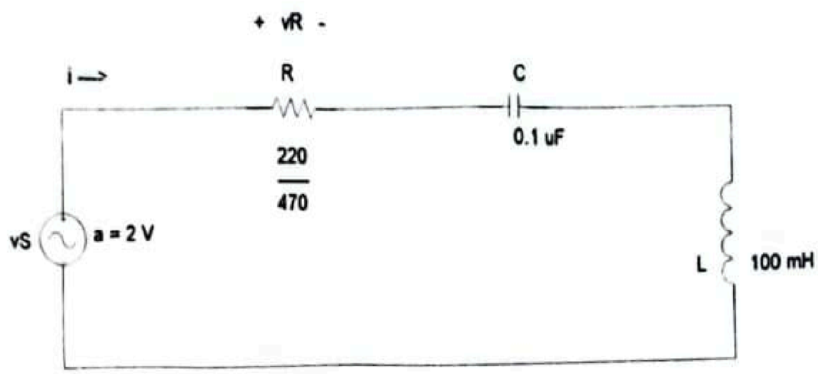
- Selectivity of a coil (Resistance & pure inductance) or series R-L-C circuit is represented by the quality factor 'Q'. The quality factor is defined as:

$$Q = \frac{CenterFrequency}{Bandwidth} = \frac{f_0}{f_1 \sim f_2} \qquad \text{.......} (2)$$

Here the center frequency $f_0 = f_r$

- From the definition of Q it is clear that for a particular f_r, the smaller the value of ($f_1 \sim f_2$), the larger the value of Q. Hence larger value of Q represents sharp peak in the resonance profile signifying that the circuit/coil is more selective to the particular frequency 'f_r'.

Circuit:



Procedure:

1. Connect the circuit as shown in the circuit diagram.
2. Adjust the voltage from the signal generator to 2 V (zero-peak).
3. Keep the frequency to some minimum value (say 10 Hz) and note the current.
4. Increase the frequency and for each frequency note the corresponding current.
5. Draw the resonant curve by taking (I) along y-axis and (f) on x-axis.
6. Find the Resonant frequency, half-power frequencies, bandwidth and Q-factor.

Data Table:

C (μF)	L (mH)	$f_{ST} = \frac{1}{2\pi\sqrt{CL}}$ (kHz)
0.1	100	

R (Ω)	V_{Sa} (V)	f_{SP} (kHz)	V_{Ramax} (V)	$0.707V_{Ramax}$ (V)	V_{Sa} (V)	f_{CL} (kHz)	V_{Sa} (V)	f_{CH} (kHz)	$BW = f_{CH} - f_{CL}$ (kHz)	$Q_S = \frac{f_{SP}}{BW}$
220	2	1.59	1.52	1.075	2	1.37	2	1.87	0.5	3.18

Saad 17/08/25

Conclusion:

In this experiment, we studied how resonance is linked with bandwidth and Q-factor. We found the resonant frequency where current was maximum, then noted the two cut off points where current dropped to 70.7% of maximum. From these values, we calculated the bandwidth and Q factor. The results showed that a higher Q gives a sharp resonance (good selectivity), while a lower Q gives a wider range.